www.RocketRevise.com

Questions

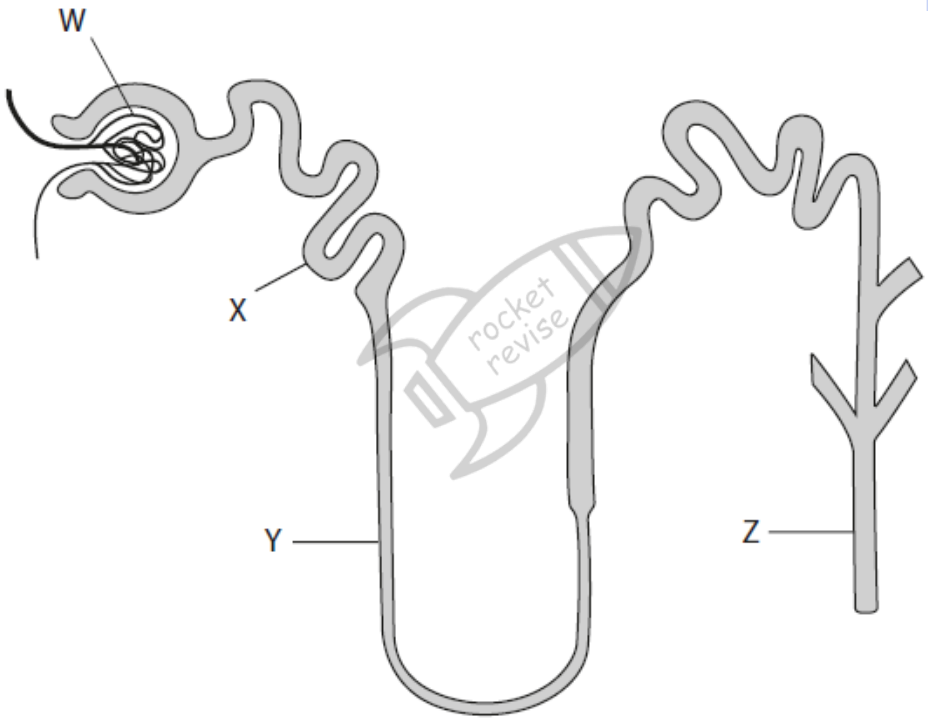
Q1.

The kidney is an organ involved in the excretion of waste materials and in the regulation of blood volume and plasma concentration.

(a) The functional unit of the kidney is the nephron.

The diagram shows a single nephron.

RocketRevise



(i) In which part of the kidney are structures W and X located?

(1)

- ☒ **A** cortex
- ☒ **B** inner medulla
- ☒ **C** outer medulla
- ☒ **D** renal pelvis

(ii) In which structure does ultrafiltration take place?

(1)

- ☒ **A** W
- ☒ **B** X

www.RocketRevise.com

www.RocketRevise.com

☒ **C** Y

☒ **D** Z

(iii) Which structures form part of the loop of Henle?

(1)

☒ **A** X only

☒ **B** X and Y

☒ **C** Y only

☒ **D** Y and Z

RocketRevise

(b) Urea is a toxic substance that is excreted from the body.

Explain how ultrafiltration removes urea from the blood.

(2)

.....

.....

.....

.....

.....

.....

(c) The kidney controls blood volume and plasma concentration by regulating the loss of water in urine.

Some mammals have adaptations to their kidneys that enable them to live in different habitats.

The table shows information about two small mammals.

RocketRevise

Mammal	Urine concentration / mOsmol dm^{-3}	Relative number of mitochondria in the thick ascending loop of Henle	Sodium ion concentration of extracellular fluid in the medulla / mmol dm^{-3}
brown rat	2900	3.6	45
kangaroo rat	5500	5.5	95

Deduce why the kangaroo rat is more successful than the brown rat at living in desert habitats.

(3)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 8 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The nervous system of an organism enables it to respond to a stimulus.

(a) A reflex action is a rapid involuntary movement in response to a stimulus. The response to a pinprick in a finger is an example of a reflex action.

(i) Which component of the nervous system continues a reflex arc immediately after the receptor has been stimulated by the stimulus?

(1)

RocketRevise

☒ **A** motor neurone

☒ **B** relay neurone

☒ **C** Schwann cell

☒ **D** sensory neurone

(ii) Which pathway shows a reflex arc?

(1)

www.RocketRevise.com

www.RocketRevise.com

☒ **A** muscle → receptor → brain

☒ **B** muscle → spinal cord → brain

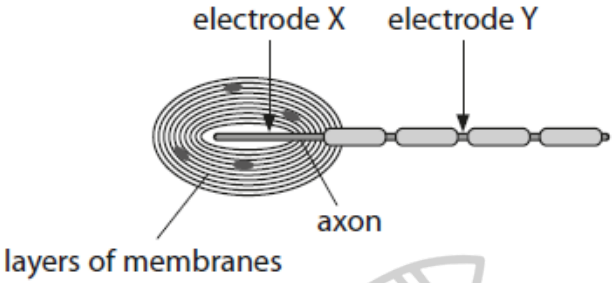
☒ **C** receptor → spinal cord → muscle

☒ **D** spinal cord → brain → muscle

(b) Pacinian corpuscles are pressure receptors found in the skin of the fingertip.

The effect of different pressures applied to the finger tip, on electrode potentials across the axon membrane of the neurone of a Pacinian corpuscle was investigated.

The diagram shows the structure of a Pacinian corpuscle and the location of electrodes, X and Y, used to measure axon membrane potentials.



RocketRevise

The table shows the results of the investigation.

Pressure applied to the fingertip	Membrane potential at electrode X / mV	Membrane potential at electrode Y / mV
None	−70	−70
Low	−60	−70
Medium	+20	+40
High	+40	+40

(i) Describe how the resting potential of −70mV is maintained in the axon when no pressure is applied.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(ii) Suggest how a change in pressure at the Pacinian corpuscle causes an action potential in the axon.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(iii) Explain why the membrane potential at electrode Y was the same when medium or high pressure was applied to the fingertip.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 11 marks)

www.RocketRevise.com

www.RocketRevise.com

Q3.

Skeletal muscle is made of bundles of fibres.

(a) Describe the role of calcium ions in producing contraction of a muscle fibre.

(2)

.....

.....

.....

.....

RocketRevise

.....

.....

(b) The table shows some features of slow and fast twitch muscle fibres.

Feature of fibre	Slow twitch fibre	Fast twitch fibre
Number of mitochondria per fibre	Many	Few
Number of myoglobin molecules per fibre	Many	Few
Capillary network	Larger	Smaller
Contraction	Slow and sustained	Rapid and intense

Olympic marathon runners have a high proportion of slow twitch fibres in their muscles.

Explain how the features of slow twitch fibres, shown in the table, are of benefit to an Olympic marathon runner.

(4)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

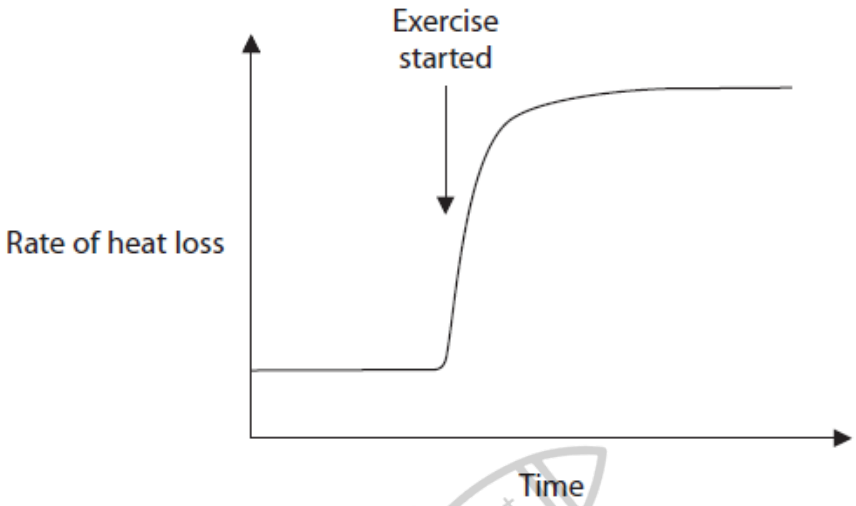
.....

.....

.....

.....

(c) The graph shows the rate of heat loss at rest and during exercise.



RocketRevise

Comment on the rate of heat loss during exercise.

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 8 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

www.RocketRevise.com

www.RocketRevise.com

Homeostasis is the state maintained by living systems.

(a) Feedback mechanisms are involved in maintaining homeostasis.

Describe the difference between a negative feedback mechanism and a positive feedback mechanism.

(2)

.....

.....

.....

.....

.....

.....

(b) Which statement about the homeostasis of thermoregulation when the human body temperature is 37 °C is correct?

(1)

☒ **A** thermoregulation is no longer required

☒ **B** the body no longer generates any heat

☒ **C** the body will be generating more heat than it can lose

☒ **D** the body temperature is maintained in a dynamic equilibrium

(c) The health of a kidney can be determined by measuring the glomerular filtration rate (GFR).

This method measures the rate at which a small molecule called creatinine is filtered from the blood.

(i) Explain how small molecules like creatinine are filtered from the blood in the glomerulus.

(3)

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

(ii) In 15 hours, a person excreted 1960 mg of creatinine in the urine.

Calculate the GFR for this person.

Give your answer to 3 significant figures.

(1)

Answer .....

(d) In type II diabetes, body cells take up less glucose in response to insulin.

RocketRevise

Many people with type II diabetes are overweight or obese.

Some of these individuals have been given surgery to help weight loss.

A clinical trial investigated the effect of surgery on the uptake of glucose by body cells in the presence of insulin.

This trial used groups of people who were non-diabetic, people in the early stage of diabetes (pre-diabetic) and people who were type II diabetics.

Group	Mean uptake of glucose / a.u.	
	Before surgery	After surgery
Non-diabetic	1.07 ± 0.29	1.39 ± 0.58
Pre-diabetic	0.55 ± 0.32	1.30 ± 0.88
Type II diabetic	0.40 ± 0.24	1.10 ± 0.87

(i) Comment on the results of this trial.

(3)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(ii) Calculate the difference that surgery makes to the mean uptake of glucose in the non-diabetic group and the type II diabetic group.

(2)

Answer .....

(Total for question = 12 marks)

RocketRevise

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

When activated, T lymphocytes can divide rapidly and secrete proteins called cytokines.

These cytokines will activate different types of lymphocyte.

(a) (i) Which is the function of tRNA in cytokine synthesis?

(1)

☒ **A** identifies an amino acid and transports it to a ribosome

☒ **B** identifies a base and transports it to a ribosome

☒ **C** forms a template for DNA polymerase

☒ **D** forms a template for RNA polymerase

RocketRevise

(ii) Which organelle is involved in chemically modifying cytokine proteins before they are secreted?

(1)

☒ **A** Golgi apparatus

☒ **B** ribosome

☒ **C** smooth endoplasmic reticulum

☒ **D** vesicle

(b) Describe the role of DNA polymerase in DNA replication during cell proliferation.

www.RocketRevise.com

www.RocketRevise.com

(2)

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

(c) Explain how cytokine genes can be expressed in an activated T lymphocyte.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(d) T lymphocytes produce different types of cytokine.

RocketRevise

Some of these cytokines activate T killer cells and some activate B cells.

Explain why different cytokines can activate different cells.

(3)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

(e) CD45 is a protein found on the surface of lymphocytes. Different forms of the CD45 protein are found on the surface of different lymphocytes.

Describe how different forms of the CD45 protein can be produced from a single CD45 gene.

(3)

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 13 marks)

Q6.

The kidneys perform important roles in filtering the blood and maintaining fluid levels in the body.

Each kidney contains up to a million functioning units called nephrons.

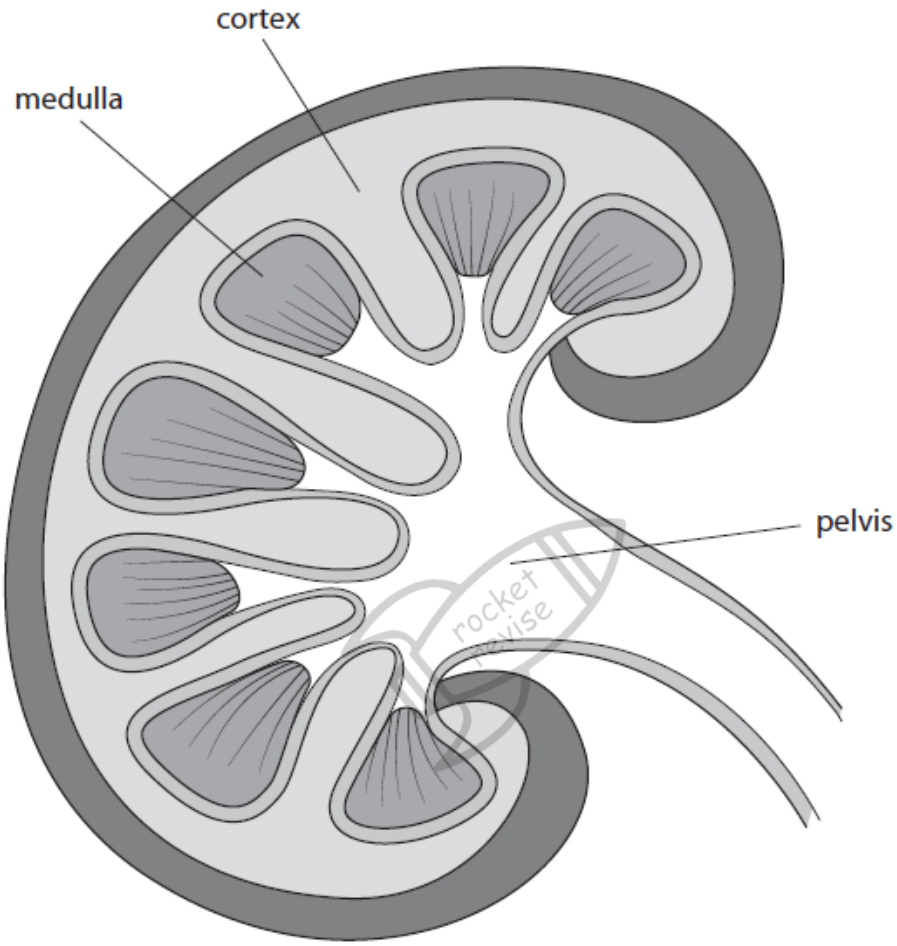
www.RocketRevise.com

www.RocketRevise.com

Each nephron consists of a filtering unit of tiny blood vessels called a glomerulus that is attached to a tubule.

(a) The diagram shows part of a human kidney.

(i) Draw an arrow on the diagram to show the location of a single Bowman's capsule (renal capsule).



RocketRevise

(1)

(ii) Where in the kidney are most amino acids reabsorbed from the nephron?

(1)

☒ **A** collecting duct

RocketRevise

☒ **B** distal tubule

☒ **C** loop of Henle

☒ **D** proximal tubule

(iii) Which row shows the correct pathway taken by a water molecule through the kidney nephron?

(1)

www.RocketRevise.com

www.RocketRevise.com

		Route through nephron →			
☒	A	collecting duct	proximal tubule	distal tubule	glomerulus
☒	B	glomerulus	proximal tubule	distal tubule	collecting duct
☒	C	glomerulus	distal tubule	collecting duct	proximal tubule
☒	D	proximal tubule	glomerulus	distal tubule	collecting duct

(b) Explain how the loop of Henle acts as a countercurrent multiplier to produce concentrated urine.

(4)

RocketRevise

(c) Describe how ADH is involved in the control of the water potential of the blood.

RocketRevise

(4)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

(Total for question = 11 marks)

RocketRevise

Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a camel.



RocketRevise

(Source: © Shengyong Li/Shutterstock)

Camels are an important livestock species. They are adapted to living in hot, dry environments.

(a) How many of the following adaptations enable a camel to reduce its water loss?

- production of very concentrated urine
- high glomerular filtration rate
- long loop of Henle
- reduced release of ADH

www.RocketRevise.com

www.RocketRevise.com

(1)

☒ **A** 0

☒ **B** 1

☒ **C** 2

☒ **D** 3

(b) The table shows the water use of five species of animal.

Species	Water use / $cm^3 kg^{-1} day^{-1}$	Mean mass / kg
buffalo	150	697
camel	57	850
goat	136	80
sheep	95	110
zebu	120	195

RocketRevise

(i) Calculate the difference in the volume of water use in a day by a buffalo and a camel.

Give your answer in dm^3 to two significant figures.

(2)

Answer ..... dm^3

(ii) Suggest two reasons why there is no correlation between water use and mean mass for the species given in the table.

(2)

.....

.....

.....

.....

.....

.....

(c) The kangaroo rat is another animal that lives in a hot, dry environment.

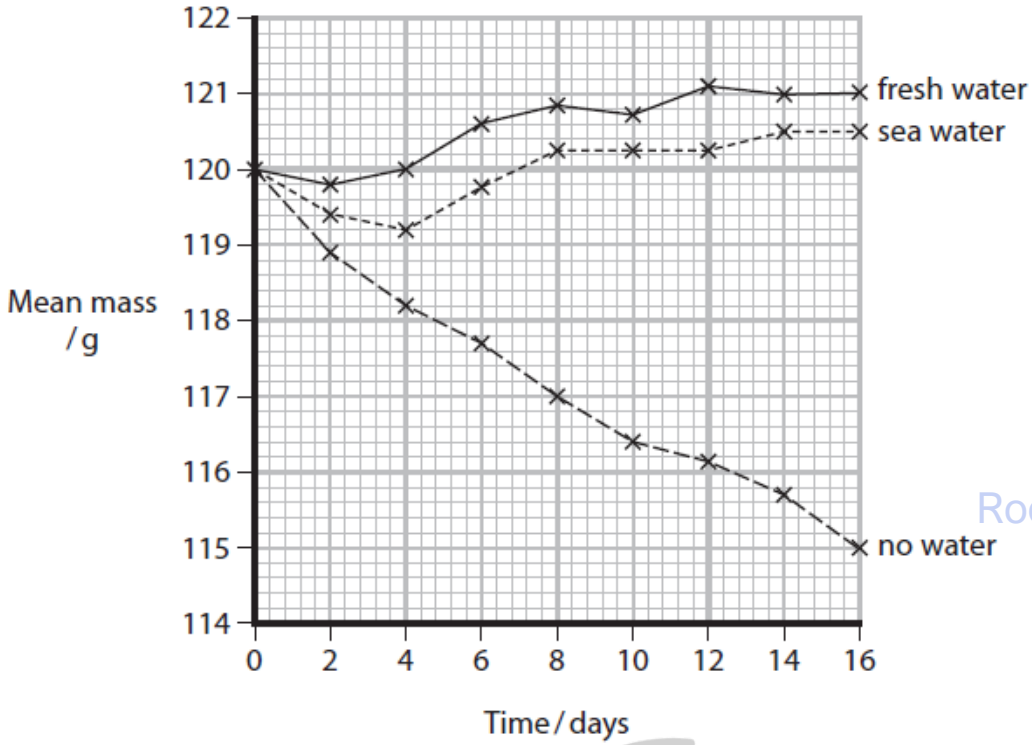
In an investigation, kangaroo rats were given either fresh water, sea water or no water.

The graph shows the mean body mass of kangaroo rats which were fed on a diet of soya beans

www.RocketRevise.com

www.RocketRevise.com

over a 16-day period.



RocketRevise

Determine the effect of the type of water given on the mean mass of the kangaroo rats in this investigation.

Use the information in the graph to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(d) The kangaroo rat's kidneys play an important role in the process of conserving water.

Explain how the nephrons (kidney tubules) in the kidneys of the kangaroo rat are able to produce very concentrated urine.

(2)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

(Total for question = 10 marks)

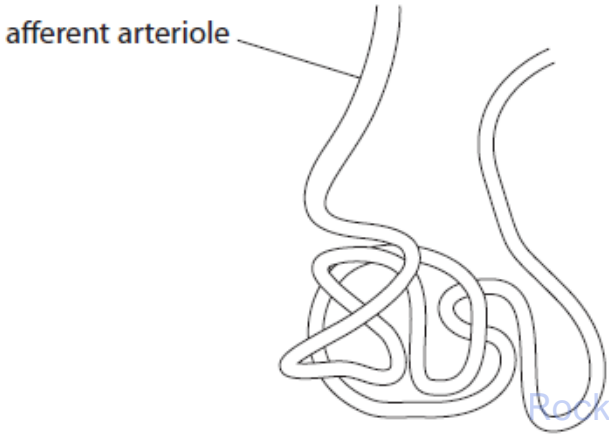
RocketRevise

Q8.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The kidney is an organ that is involved in the removal of waste metabolites from the blood.

(a) The drawing shows the blood supply inside a Bowman's capsule (renal capsule).



RocketRevise

(i) Complete the diagram to show the Bowman's capsule and label two structures in this diagram.

(3)

(ii) Which substances are normally filtered from the blood in the Bowman's capsule?

(1)

☒ **A** amylose and glucose

☒ **B** glucose and sodium chloride

☒ **C** potassium ions and haemoglobin

www.RocketRevise.com

www.RocketRevise.com

☒ **D** urea and prothrombin

(iii) Which row shows the route taken by a water molecule through a nephron in the kidney?

(1)

		Route through a nephron →			
☒	**A**	collecting duct	proximal tubule	distal tubule	loop of Henle
☒	**B**	loop of Henle	proximal tubule	collecting duct	distal tubule
☒	**C**	proximal tubule	distal tubule	loop of Henle	collecting duct
☒	**D**	proximal tubule	loop of Henle	distal tubule	collecting duct

RocketRevise

(b) Describe how glucose reabsorption takes place in the nephron.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) Urea is produced by the metabolic breakdown of some organic molecules.

Which statement describes the production of urea?

RocketRevise

(1)

☒ **A** in the kidneys from excess amino acids

☒ **B** in the kidneys from glycogen

☒ **C** in the liver from excess amino acids

☒ **D** in the liver from glycogen

(Total for question = 9 marks)

www.RocketRevise.com

www.RocketRevise.com

Q9.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When antagonistic muscles contract, they move skeletal bones.

(a) (i) Energy is needed when a muscle contracts.

RocketRevise

Which row in the table shows the energy use in muscle contraction?

(1)

		Energy in muscle contraction is used to		
		break the myosin–actin cross bridge	remove calcium ions from the sarcoplasm	release ADP and Pi from ATP
☒	A	✓	✓	✓
☒	B	✓	✓	✗
☒	C	✗	✗	✓
☒	D	✗	✗	✗

(ii) Describe the sliding filament theory of muscle contraction.

(4)

.....

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

(b) Two types of ostrich, the South African Black and the Zimbabwean Blue, are farmed in South Africa.

Ostrich muscle is a source of dietary protein.

Scientists compared the mean masses of three muscles from the ostriches.

Muscle	Mean mass of muscle and standard deviation / kg	
	South African Black (n = 34)	Zimbabwean Blue (n = 2)
T	0.69 ± 0.10	0.88 ± 0.11
G	0.59 ± 0.08	0.74 ± 0.03
F	0.40 ± 0.06	0.50 ± 0.01

RocketRevise

(i) State **two** conclusions that can be drawn about the mean mass of muscles shown in the table.

(2)

.....

.....

.....

.....

.....

.....

(ii) Comment on the validity of conclusions drawn from the data for these two groups of ostrich.

RocketRevise

(2)

.....

.....

.....

.....

.....

.....

(iii) Calculate the percentage difference in the mean mass of muscle G of the South African Black and the Zimbabwean Blue ostriches.

www.RocketRevise.com

www.RocketRevise.com

Use the equation: percentage difference between a and $b = \frac{a-b}{\left(\frac{a+b}{2}\right)} \times 100$

Give your answer to **two** decimal places.

(2)

Answer ..... %

(c) The hypothalamus is involved in maintaining a steady internal temperature in the ostrich.

Describe the role of the hypothalamus in maintaining a steady internal temperature.

(3)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 14 marks)

RocketRevise

Q10.

The kidney is an organ that is involved in osmoregulation and the production of urine.

(a) (i) Which substances are filtered from the blood in the Bowman's (renal) capsule of a healthy individual?

(1)

☒ **A** glucose and prothrombin

☒

www.RocketRevise.com

www.RocketRevise.com

B glucose and urea

☒ **C** glycogen and urea

☒ **D** urea and prothrombin

(ii) Which part of the Bowman's (renal) capsule prevents plasma proteins from being filtered out of the bloodstream?

(1)

☒ **A** basement membrane

☒ **B** smooth muscle of the capillaries

☒ **C** epithelial cells of the collecting duct

RocketRevise

☒ **D** epithelial cells of the loop of Henlé

(iii) Which transport mechanism is responsible for the uptake of glucose into the cells of the proximal tubule?

(1)

☒ **A** endocytosis

☒ **B** exocytosis

☒ **C** osmosis

☒ **D** sodium co-transport

(b) Describe how the loop of Henlé acts as a countercurrent multiplier.

(4)

.....

.....

.....RocketRevise.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

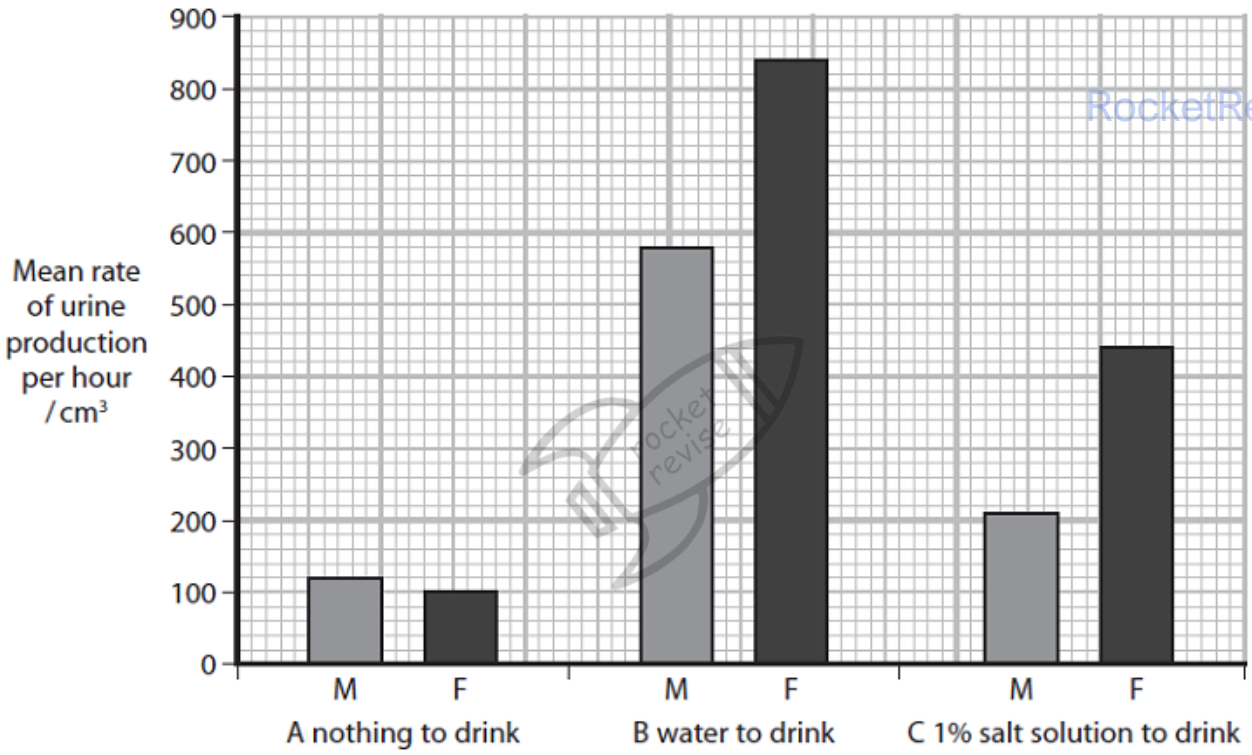
.....

.....

(c) In an investigation, one group of people was given nothing to drink, a second group was given water and the third group was given 1% salt solution to drink.

The mean rate of urine production was recorded for the males and females in each group.

The graph shows the results of this investigation.



(i) Calculate the ratio of the mean rate of urine production for the females in the three groups.

Express your answer as nothing to drink : water to drink : salt solution to drink.

Give your answer to one significant figure.

(2)

Answer .....

(ii) Describe **two** conclusions that can be drawn from the results of this investigation.

(2)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

(iii) Explain the effect on the mean urine output of the group drinking water.

(2)

.....

RocketRevise

.....

.....

.....

.....

.....

(Total for question = 13 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

RocketRevise

Cancers can develop as a result of mutations or epigenetic modifications.

(a) (i) State what is meant by the term **mutation**.

(1)

.....

.....

.....

(ii) Name the type of nuclear division taking place as a tumour develops.

(1)

www.RocketRevise.com

www.RocketRevise.com

.....

(b) How many of the following statements about epigenetic modification are correct?

(1)

- It can result in a change in the base sequence of a gene
- It can involve histone modification
- It can result in an altered phenotype
- It can involve DNA methylation

RocketRevise

☒ **A** 1

☒ **B** 2

☒ **C** 3

☒ **D** 4

(c) Oestrogen stimulates some types of breast cancer to proliferate.

Oestrogen is a steroid hormone found in mammals.

Oestrogen affects up to 100 different genes.

Explain how oestrogen may result in the proliferation of some breast cancer cells.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

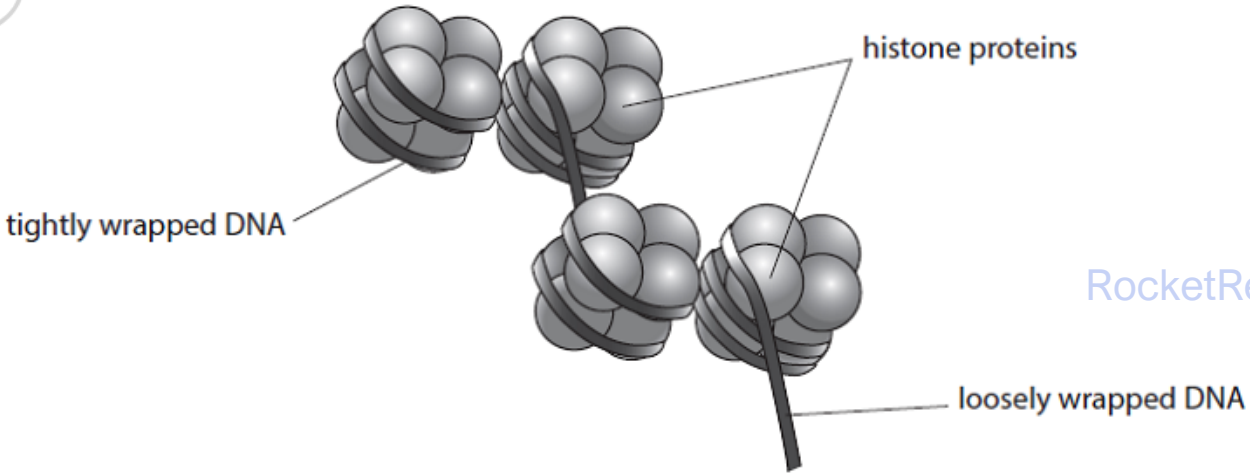
www.RocketRevise.com

.....

(d) The expression of genes in cancer cells can be regulated by histone modification.

DNA that is not being transcribed is tightly wrapped around histone proteins.

The diagram shows DNA wrapped around histone proteins.



RocketRevise

Acetylation of histones is one example of histone modification. Acetyl groups are added to histone proteins.

Suggest how acetylation of histone proteins could result in increased gene expression.

Use the information in the diagram to support your answer.

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 9 marks)

Q12.

The kidney is involved in osmoregulation and the production of urine.

www.RocketRevise.com

www.RocketRevise.com

(a) (i) Which substances are filtered from the blood in the renal capsule?

(1)

☒ **A** glucose and fibrinogen

☒ **B** glucose, fibrinogen and urea

☒ **C** glucose and urea

☒ **D** urea and prothrombin

(ii) Which structure is acting as the filter in the renal capsule?

(1)

RocketRevise

☒ **A** basement membrane

☒ **B** cells lining the renal capsule

☒ **C** endothelial cells in the walls of the capillaries

☒ **D** podocytes

(iii) Which transport mechanism is responsible for the uptake of glucose into the cells of the wall of the proximal tubule?

(1)

☒ **A** diffusion

☒ **B** endocytosis

☒ **C** sodium co-transport

☒ **D** osmosis

(iv) Explain why some urea is reabsorbed in the proximal tubule.

(2)

RocketRevise

.....

.....

.....

.....

.....

.....

(b) In an investigation, salt solution was injected into the carotid artery of a mammal.

The carotid artery carries blood to the head.

www.RocketRevise.com

www.RocketRevise.com

The rate of urine production was measured after the injection.

The table shows the results.

Time after injection of salt solution / min	Rate of urine production / $cm^3\ min^{-1}$
0	7.0
5	1.1
10	0.6
15	1.0
20	1.5
25	2.2
30	2.4

RocketRevise

(i) Calculate the percentage decrease in the rate of urine production 10 minutes after the injection.

Give your answer to 2 decimal places.

(2)

Answer ..... %

(ii) At what times should more measurements be taken to find the lowest rate of urine production?

(1)

☒ **A** at 1-minute intervals between 5 minutes and 10 minutes

☒ **B** at 1-minute intervals between 5 minutes and 15 minutes

☒ **C** at 1-minute intervals between 10 minutes and 15 minutes

RocketRevise

☒ **D** at 5 minutes, 10 minutes and 15 minutes, two more times

(iii) Explain the results of this investigation. Use the information in the table to support your answer.

(4)

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 12 marks)

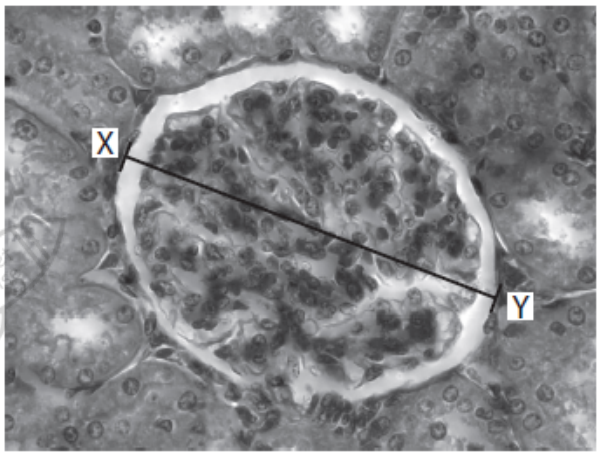
Q13.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The kidney has an important role in the regulation of the plasma concentration and blood volume.

The photograph shows a Bowman's capsule (renal capsule).

The line from X to Y shows the diameter of the Bowman's capsule.



RocketRevise

(Source: © Science Photo Library/Alamy Stock Photo)

(a) The diameter from X to Y is 70 nm.

(i) Calculate the magnification of this photograph.

Give your answer in **standard form**.

www.RocketRevise.com

www.RocketRevise.com

(2)

Answer .....

(ii) Calculate the volume of this Bowman's capsule.

Assume that this Bowman's capsule is spherical.

Use the equation:

$$V = \frac{4}{3}\pi r^3$$

Give your answer to **two** significant figures.

(2)

RocketRevise

Answer ..... nm^3

(b) (i) Which is the specialised receptor that detects changes in the plasma solute concentration?

(1)

☒ **A** baroreceptor

☒ **B** chemoreceptor

☒ **C** osmoreceptor

☒ **D** thermoreceptor

(ii) Suggest **one** reason why the plasma solute concentration of the blood needs to be kept within a narrow range.

(1)

.....

.....

RocketRevise

.....

(c) Drinking too much water causes the plasma solute concentration to decrease and the blood volume to increase.

Describe how the body responds to this change.

(4)

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(d) (i) Which method reabsorbs glucose from the proximal convoluted tubule into the blood?

(1)

☒ **A** co-transport with sodium ions

☒ **B** endocytosis

☒ **C** exocytosis

☒ **D** osmosis

(ii) The Bowman's capsule of a healthy kidney filters up to 7.5 g of glucose per hour.

Calculate the rate of glucose filtration in **milligrams per second**.

(2)

Answer ..... mg s^{-1}

(Total for question = 13 marks)

RocketRevise

Q14.

The kangaroo rat is a mammal adapted to living in a warm dry habitat.

(a) The resting body temperature of a kangaroo rat is 37 °C.

During exercise the body temperature is maintained between 38.5 °C and 40 °C.

Explain how the body temperature can be regulated to stay between 38.5 °C and 40 °C.

www.RocketRevise.com

www.RocketRevise.com

(4)

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

.....

(b) In mammals, urine is produced by ultrafiltration and the selective reabsorption of solutes and water.

Kangaroo rats are adapted to survive on a diet of dry seeds and little or no drinking water.

Suggest how kangaroo rats are adapted to survive in areas with little drinking water and dry seeds.

(3)

.....

.....

.....

RocketRevise

.....

.....

.....

.....

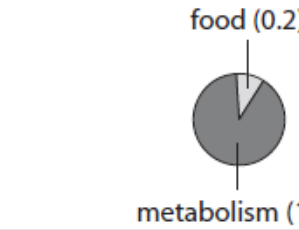
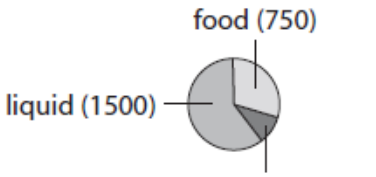
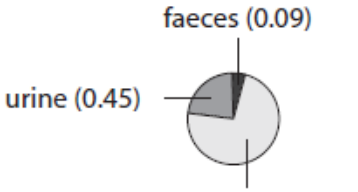
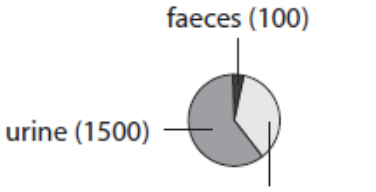
.....

.....

(c) The diagram compares the daily water balance between a kangaroo rat and a human.

www.RocketRevise.com

www.RocketRevise.com

	Water balance in a kangaroo rat (2 cm^3 / day)	Water balance in a human (2500 cm^3 / day)
Water gain / cm^3		
Water loss / cm^3		

RocketRevise

(i) Calculate the difference in urine produced over seven days by the kangaroo rat and by the human.

Give your answer in standard form.

(2)

Answer .....

(ii) Comment on the differences in water gain and water loss between the kangaroo rat and the human.

Use your knowledge and information in the diagram to support and explain your answer.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 13 marks)

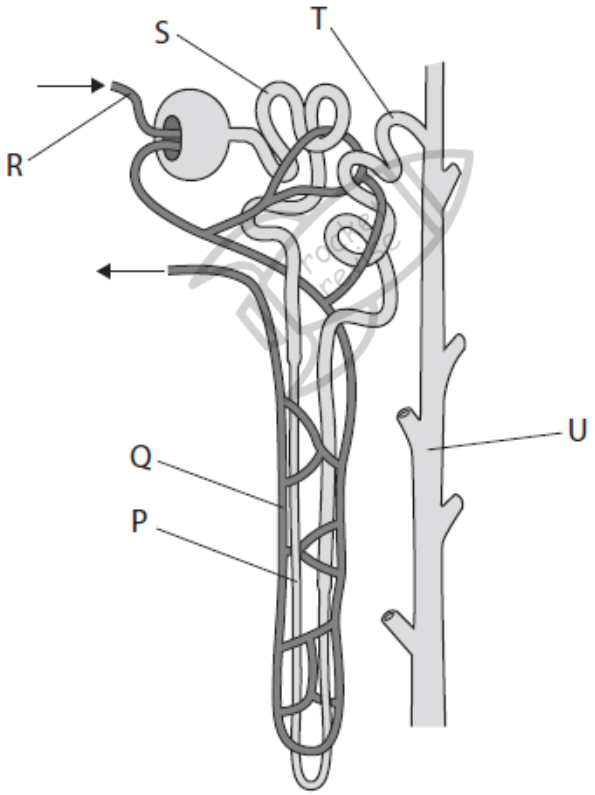
Q15.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The kidney is involved in the reabsorption of solutes and water.

RocketRevise

(a) The diagram shows a mammalian kidney tubule (nephron) with its blood supply.



(i) Which labelled structure transports blood to the renal vein?

RocketRevise

(1)

☒ **A** Q

☒ **B** R

☒ **C** S

☒ **D** U

(ii) Which labelled structures would be found in the medulla region of the kidney?

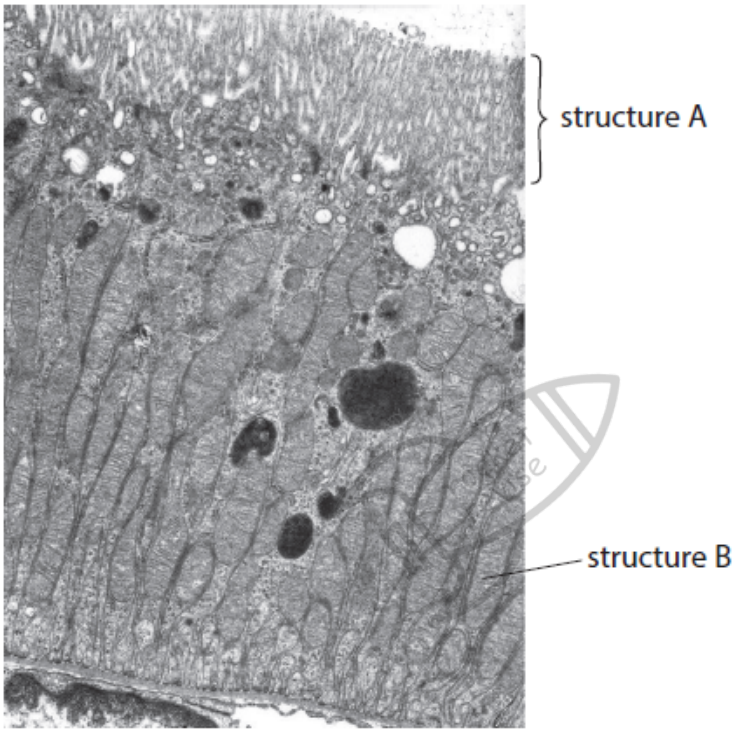
(1)

www.RocketRevise.com

www.RocketRevise.com

		Structures		
☒	**A**	P	Q	U
☒	**B**	R	Q	P
☒	**C**	R	S	T
☒	**D**	S	T	U

(iii) The electron micrograph shows part of a cell lining the proximal convoluted tubule.



RocketRevise

(Source: © Jose Calvo/Science Photo Library)

Explain how structures A and B and their locations in the proximal convoluted tubule enable the reabsorption of solutes.

(3)

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

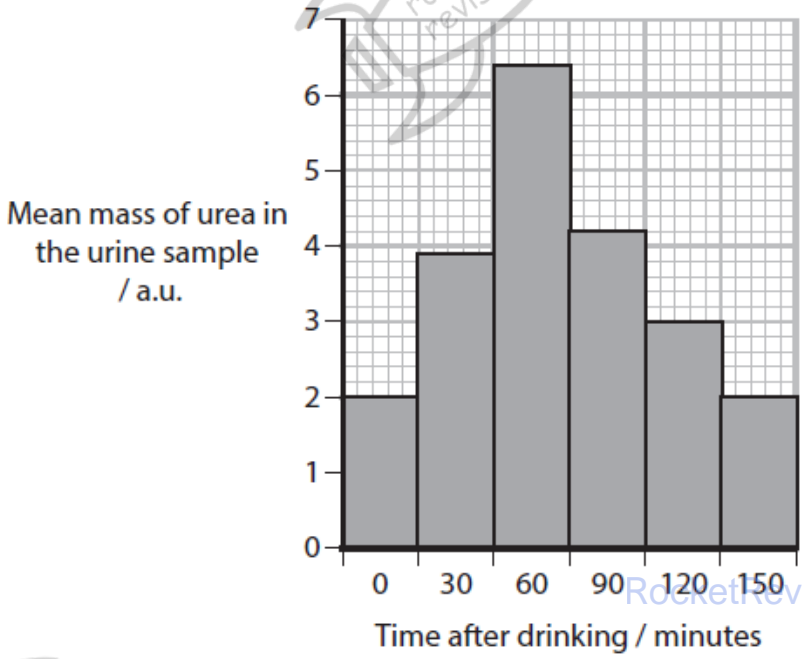
(b) A group of individuals gave a urine sample and then immediately drank 1 000 cm^3 of water.

Urine samples were then collected at 30-minute intervals.

The volumes and composition of urine produced were then recorded.

The table and graph show the results of this investigation.

Time after drinking / minutes	Mean volume of urine produced / cm^3	Mean concentration of ions in urine / a.u.
0	100	6.0
30	195	3.0
60	325	1.8
90	210	2.8
120	150	4.0
150	100	6.0



Comment on the results of this investigation.

Use the information in the table and the graph and your own knowledge to support your answer.

(4)

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(c) A person with hyponatraemia suffers from excessive production of antidiuretic hormone (ADH) from the
pituitary gland.

This causes changes in the sodium ion concentration of the blood.

The table shows the sodium ion concentration of the blood in patients with hyponatraemia.

Condition	Sodium ion concentration of the blood / a.u.
normal	135 to 145
mild hyponatraemia	130 to 134
moderate hyponatraemia	125 to 129
severe hyponatraemia	<125

Explain how the excessive release of antidiuretic hormone (ADH) causes these changes in blood sodium concentrations.

(3)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(Total for question = 12 marks)

Q16.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The kidney is an organ in the human body that is involved in blood filtration and the production of urine.

(a) (i) The afferent arteriole leading to the glomerulus is wider than the efferent arteriole leaving the glomerulus.

Which describes the effect that this produces?

(1)

☐ **A** it increases pressure in the glomerulus for ultrafiltration

☐ **B** it increases water movement by osmosis into the Bowman's (renal) capsule

☐ **C** it reduces the risk of capillary damage, due to high pressure

☐ **D** it reduces water reabsorption from the proximal tubule

(ii) In each hour, the kidneys will filter 6 000 cm^3 of blood.

How much blood will be filtered in 24 hours?

(1)

☐ **A** 124 dm^3

☐ **B** 144 cm^3

☐ **C** 144 dm^3

☐ **D** 1 440 cm^3

(b) The table shows the relationship between the ADH concentration in blood, concentration of urine and the rate of urine production.

www.RocketRevise.com

www.RocketRevise.com

Percentage of maximum ADH concentration in blood (%)	Concentration of urine / a.u.	Rate of urine production / cm^3 $minute^{-1}$
0	90	20.0
25	700	8.0
50	950	4.0
75	1050	1.5
100	1150	1.0

(i) Calculate the volume of urine produced in **one** day with an ADH concentration of 75%.

RocketRevise

Assume that the concentration of ADH stays constant throughout the day.

Give your answer in standard form.

(2)

Answer..... cm^3

(ii) Describe the effect of ADH concentration on the concentration of urine and the rate of urine production.

(2)

.....

.....

.....

.....

.....

.....

(c) Explain how ADH affects the movement of water from the lumen of the collecting duct into the blood.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

(Total for question = 9 marks)

RocketRevise

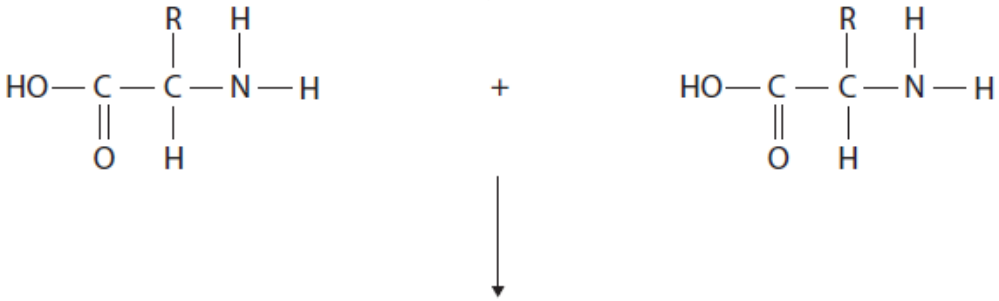
Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Proteins are one type of biological molecule.

(a) (i) Complete the diagram to show the molecules formed when these two amino acids join together.

(2)



(ii) Which is the type of reaction that joins these two molecules together?

(1)

RocketRevise

☒ **A** condensation

☒ **B** decarboxylation

☒ **C** hydrolysis

☒ **D** phosphorylation

(b) Excess proteins are broken down into amino acids.

Describe what happens to these amino acids.

(3)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

(Total for question = 6 marks)

RocketRevise

www.RocketRevise.com